

Myocardial Perfusion Segmentation with PyTorch U-Net

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Abstract. This study investigates deep learning-based segmentation of myocardial perfusion regions from medical imaging data using a U-Net architecture. The dataset consists of volumetric DICOM image series paired with NIfTI segmentation masks, which are preprocessed into two-dimensional slices and normalized for model input. A baseline U-Net model with a pretrained ResNet34 encoder achieved strong performance, reaching a validation Dice coefficient of approximately 0.86. A tuned variant incorporating data augmentation, AdamW optimization, and learning rate scheduling was also evaluated. Despite increased model complexity, the tuned approach underperformed, highlighting the sensitivity of segmentation models to small datasets and class imbalance. The results demonstrate that careful preprocessing and appropriate model simplicity are critical factors in achieving robust segmentation performance.

Keywords: Medical Image Segmentation · U-Net · Deep Learning · Dice Coefficient · DICOM

1 Introduction

Medical image segmentation is a critical task in biomedical analysis, enabling the extraction of anatomical structures from imaging modalities such as MRI and CT scans. Given an input image $X \in R^{H \times W}$, the objective is to learn a mapping $f_\theta : X \rightarrow \hat{Y}$, where $\hat{Y} \in \{0, 1\}^{H \times W}$ represents a binary segmentation mask.

U-Net architectures have become a standard approach due to their encoder-decoder structure and skip connections, which allow simultaneous capture of global context and fine spatial details. However, segmentation tasks in medical imaging are often complicated by limited data availability and strong class imbalance, where the foreground region occupies only a small portion of the image.

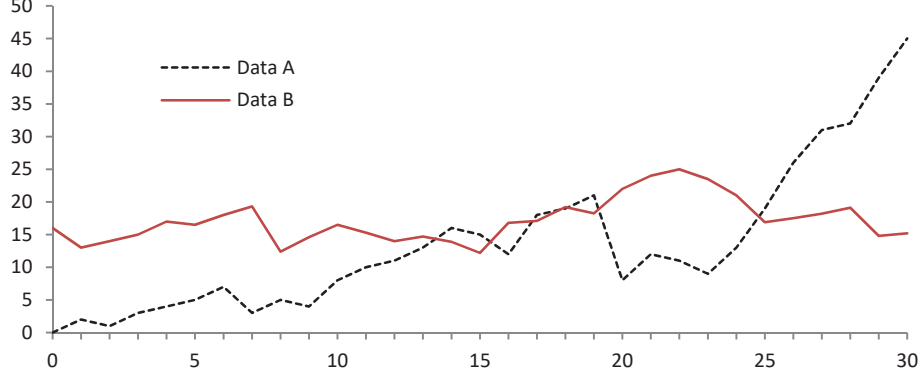


Fig. 1. Example DICOM slice, corresponding ground truth mask, and overlay visualization demonstrating spatial alignment.

2 Data and Preprocessing

The dataset consists of volumetric DICOM image series paired with NIfTI masks. Each volume is decomposed into a set of two-dimensional slices $\{X_i\}_{i=1}^N$ with corresponding binary masks $\{Y_i\}_{i=1}^N$.

To standardize input data, each image is normalized using min-max scaling:

$$X'_i = \frac{X_i - \min(X_i)}{\max(X_i) - \min(X_i) + \epsilon} \quad (1)$$

Binary masks are defined such that:

$$Y_i(p) = \begin{cases} 1, & \text{if } p \in \Omega_i \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where Ω_i denotes the set of pixels belonging to the region of interest in slice i .

A significant portion of slices contains no foreground pixels:

$$\sum_p Y_i(p) = 0 \quad (3)$$

To address this imbalance, stratified splitting is applied to maintain consistent proportions of empty and non-empty masks across training, validation, and test sets.

3 Model Architecture

The segmentation model is based on the U-Net architecture, consisting of an encoder and decoder with skip connections. The encoder extracts hierarchical features:

$$Z_l = \phi_l(Z_{l-1}), \quad l = 1, \dots, L \quad (4)$$

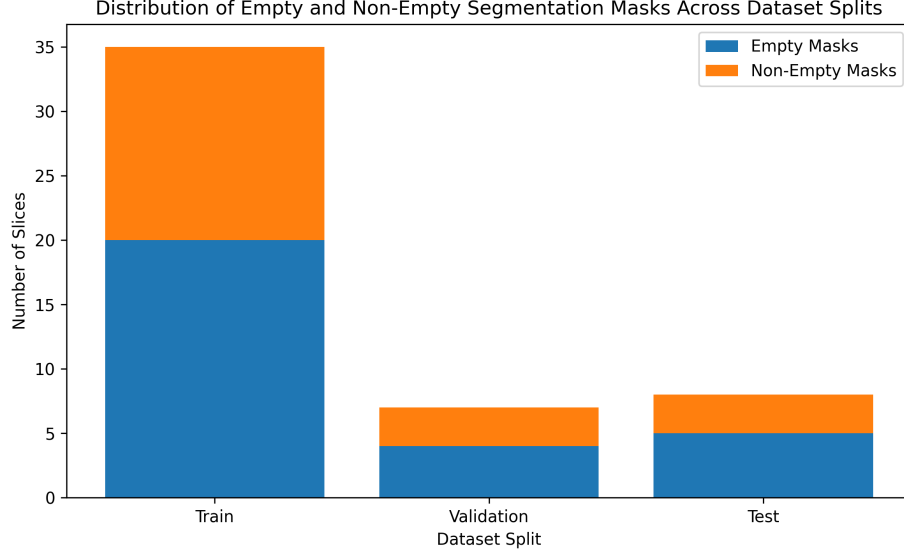


Fig. 2. Distribution of empty and non-empty segmentation masks across dataset splits.

A pretrained ResNet34 model serves as the encoder, providing feature representations learned from large-scale datasets. The decoder reconstructs the segmentation mask through upsampling and feature fusion:

$$\hat{Y} = f_{\theta}(X) \quad (5)$$

Skip connections between encoder and decoder layers preserve spatial information, improving segmentation accuracy.

4 Loss Function and Evaluation

The model is trained using a combined loss function consisting of binary cross-entropy and Dice loss. The binary cross-entropy loss is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N \left[Y_i \log(\hat{Y}_i) + (1 - Y_i) \log(1 - \hat{Y}_i) \right] \quad (6)$$

The Dice coefficient measures overlap between predicted and ground truth masks:

$$\text{Dice} = \frac{2 \sum_p \hat{Y}(p) Y(p)}{\sum_p \hat{Y}(p) + \sum_p Y(p) + \epsilon} \quad (7)$$

The Dice loss is defined as:

$$\mathcal{L}_{Dice} = 1 - \text{Dice} \quad (8)$$

The final loss function is:

$$\mathcal{L} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} \quad (9)$$

5 Experimental Setup

Two models were evaluated: a baseline U-Net and a tuned variant. The baseline model uses a pretrained ResNet34 encoder and standard normalization. The tuned model incorporates data augmentation and advanced optimization strategies.

Data augmentation is defined as a stochastic transformation:

$$X' = T(X) \quad (10)$$

where T represents operations such as rotation and flipping.

Model parameters are optimized using gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L} \quad (11)$$

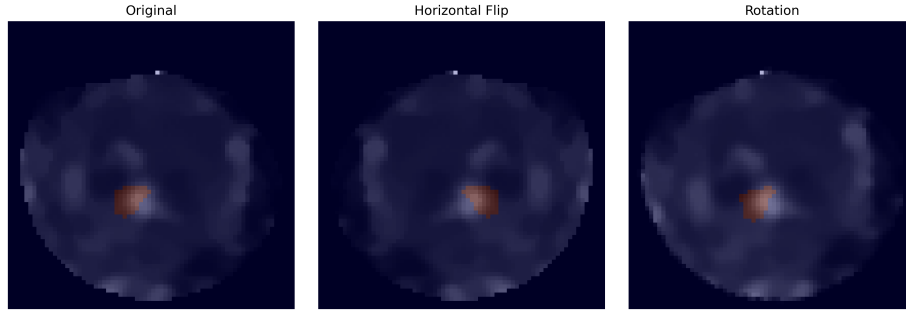


Fig. 3. Examples of augmented images demonstrating spatial transformations applied during training.

6 Results

The baseline model achieved a validation Dice coefficient of approximately 0.86, indicating strong segmentation performance. In contrast, the tuned model achieved significantly lower performance, with validation Dice scores near 0.24.

Analysis of predictions revealed that the tuned model often over-segmented, producing large false-positive regions:

$$\sum_p \hat{Y}(p) \gg \sum_p Y(p) \quad (12)$$

This behavior reduces overlap and leads to lower Dice scores despite moderate loss values.

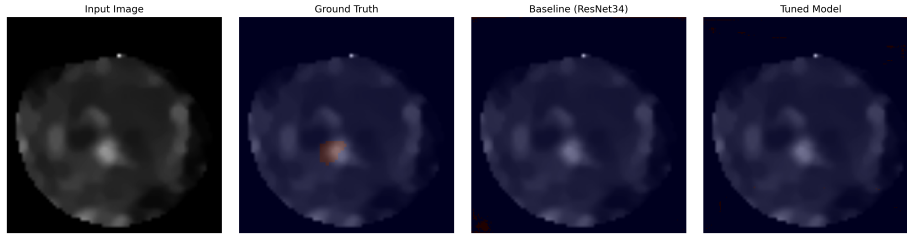


Fig. 4. Comparison of baseline and tuned model predictions alongside ground truth masks.

7 Discussion

The results demonstrate that segmentation performance is highly sensitive to dataset size, class imbalance, and augmentation strategies. While augmentation is typically beneficial, excessive transformations can disrupt spatial consistency, particularly in small datasets.

The success of the baseline model highlights the effectiveness of pretrained feature extractors. The ResNet34 encoder provides robust feature representations that generalize well even with limited training data.

8 Conclusion

This study presents a deep learning-based approach for myocardial perfusion segmentation using a U-Net architecture. The baseline model achieved strong performance, while a tuned variant incorporating additional complexity did not improve results. These findings suggest that, in small and imbalanced datasets, simpler models with careful preprocessing can outperform more complex configurations. Future work may explore larger datasets, three-dimensional models, and specialized loss functions for improved segmentation performance.

References

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